

Your personal quantum computer?

In the realm of scientific discovery, few pursuits have been as tantalising and elusive as the creation of a room-temperature, ambient-pressure superconductor. Its advent could revolutionise our technological landscape, from consumer electronics to quantum computing, and even energy sustainability. Enter LK-99, a compound that may just be the game-changer we've been waiting for. It has the potential to democratise technological innovations that would otherwise have not left the lab for at least 15-20 more years.

Superconductors have been the metaphorical Holy Grail of material science. Imagine a world where your high-performance CPU operates without power losses, where supercomputers hum with almost perfect efficiency, and where electric motors and generators function with unprecedented effectiveness. The environmental implications alone are monumental. The journey to this has been a rollercoaster ride of scientific intrigue, with the initial discovery of LK-99 by a Korean team sparking a global race to validate its properties. Preliminary confirmations from two separate research teams, one from the U.S.'s Lawrence Berkeley National Lab and another from China's Huazhong University of Science and Technology, have added fuel to the fire.

LK-99, a complex compound resulting from a multi-step synthesis process involving Lanarkite (Pb_2SO_5) and Copper Phosphide (Cu_3P), has been the subject of intense study. Simulations suggest that the material should manifest superconduction pathways for electrons to travel through unimpeded and without resistance. These pathways, intriguingly, only form in the highest-energy areas of the resulting crystal lattice. The Chinese team went a step further, claiming to have successfully replicated the superconductor's manufacturing process and posting a video showcasing the Meissner effect, a phenomenon associated with superconductivity, as proof. The video has been hailed as a significant piece of evidence supporting the superconducting capabilities of LK-99. However, the journey to validation is



“Superconductors, materials that conduct electricity without any losses, have been the metaphorical Holy Grail of material science.”

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far from over. The original published articles do not claim to have seen definitive features of superconductivity. Instead, they show the material exhibiting strong diamagnetic properties, including partial levitation on top of a large magnet, which is correlated with superconductivity. What makes this material even more intriguing is that it is based on a novel theory. Superconductors until now have been explained by the standard Bardeen-Cooper-Schrieffer (BCS) theory of superconductivity. So not only is this new material making tall claims but it is also based on an approach that is yet to be accepted by the mainstream scientific community. The proposed mechanism for superconductivity in LK-99 is based on a novel “BR-BCS” theory of superconductivity, which combines a classical theory of metal-insulator transitions with the standard Bardeen-Cooper-Schrieffer theory of superconductivity. This theory, along with ideas from the controversial theory of hole superconductivity by J.E.Hirsch, forms the basis for the claims made by the Korean team. You can see a lot of reasons to doubt their claims.

Every passing day sees a portion of the findings getting replicated or failing replication. As of August 1, 2023, none of the expected properties of high-temperature superconductivity have been observed by the original experiment or attempted replications. But a mere day later, researchers from Huazhong University of Science and Technology have replicated the manufacturing process and posted video proof. There's just too much excitement and by the time you read this, we might already know this year's Nobel winners.

Despite the scepticism, if the claims are proven to be true, this material could represent one of humanity's major technological milestones. The world watches with bated breath as the scientific community continues to unravel the mysteries of LK-99. The road to validation may be long and fraught with challenges, but the potential rewards are too great to ignore and LK-99 may just be the breakthrough we've been waiting for. ■